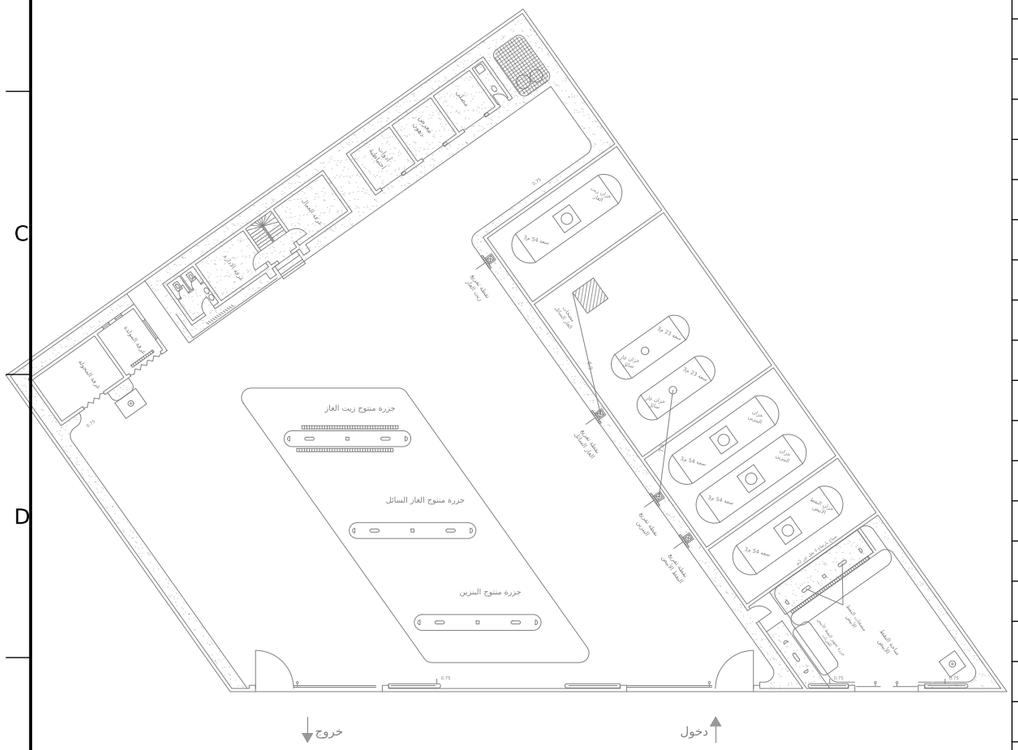
FUEL FILLING STATION

محطة تعبئة وقود - القطعة المرقمة (1 / 3453) مقاطعة (43) يارمجة الشرقية
قضاء الموصل / محافظة نينوى - للمواطنين (هادي حسين جمعة - احمد عثمان عيد)

ELECTRICAL - CIVIL DEFENCE - SANITARY - STRUCTURAL

DESIGN DRAWINGS, ISSUE 00 (STAGE 0: GENERAL SHEETS)

OWNERS: HADI HUSSEIN JUMAA - AHMED OTHMAN EID



SITE PLAN (REFERENCE - SEE FUEL_STATION_SITE_PLAN.dxf)
PLOT AREA 2,415.25 m2 - STREET FRONT 61.28 m
ABOVE-GROUND STORAGE 262 m3 IN 6 TANKS - 9 DISPENSERS

DRAWING REGISTER

No.	DRAWING	TITLE	STATUS
01	G-001	COVER SHEET - DRAWING REGISTER	ISSUED FOR REVIEW
02	G-002	LEGEND - SYMBOLS AND LINE TYPES	ISSUED FOR REVIEW
03	G-003	DESIGN BASIS - CODES, DATA AND GENERAL NOTES	ISSUED FOR REVIEW
04	F-001	SAFETY DISTANCES - SEPARATION PLAN AND COMPLIANCE TABLE	PLANNED - STAGE 1
05	F-002	FIRE WATER NETWORK - HYDRANTS, HOSE REELS, LPG SPRAY COOLING	PLANNED - STAGE 1
06	F-003	PORTABLE EXTINGUISHERS - LAYOUT AND QUANTITY SCHEDULE	PLANNED - STAGE 1
07	F-004	FIRE ALARM AND ESD - DETECTION, CALL POINTS, SHUT-DOWN	PLANNED - STAGE 1
08	F-005	EVACUATION PLAN - EXITS, ROUTES, TANKER CIRCULATION	PLANNED - STAGE 1
09	F-006	FIRE WATER SCHEMATIC - PUMP SET AND RISER DIAGRAM	PLANNED - STAGE 1
10	F-007	BUND COMPARTMENTS - CAPACITY CHECK, DRAIN VALVES, SUMPS	PLANNED - STAGE 1
11	F-008	CIVIL DEFENCE DETAILS - SIGNS, CABINETS, MOUNTING	PLANNED - STAGE 1
12	E-001	POWER SUPPLY ARRANGEMENT - TRANSFORMER, GENERATOR, MDB, ATS	PLANNED - STAGE 2
13	E-002	HAZARDOUS AREAS - ZONE 0/1/2 CLASSIFICATION AND SECTIONS	PLANNED - STAGE 2
14	E-003	EXTERNAL POWER - DISPENSER AND TANK CABLES, TRENCHES	PLANNED - STAGE 2
15	E-004	EXTERNAL LIGHTING - YARD AND BOUNDARY (CANOPY IN STAGE 6)	PLANNED - STAGE 2
16	E-005	BUILDINGS LIGHTING - LAYOUT AND CIRCUITS	PLANNED - STAGE 2
17	E-006	BUILDINGS POWER - SOCKETS, A/C AND CIRCUITS	PLANNED - STAGE 2
18	E-007	SINGLE LINE DIAGRAM - MDB, SMDBs, ATS AND ESD	PLANNED - STAGE 2
19	E-008	LOAD SCHEDULES - MAXIMUM DEMAND AND PANEL SCHEDULES	PLANNED - STAGE 2
20	E-009	EARTHING AND BONDING - EARTH NETWORK AND TANKER STATIC EARTH	PLANNED - STAGE 2
21	E-010	LIGHTNING PROTECTION - BUILDINGS AND TANKS (CANOPY IN STAGE 6)	PLANNED - STAGE 2
22	P-001	WATER SUPPLY - MAINS, GROUND TANK, PUMP, DISTRIBUTION	PLANNED - STAGE 3
23	P-002	FOUL DRAINAGE - MANHOLES, SEPTIC TANK, SOAKAWAY	PLANNED - STAGE 3
24	P-003	OILY AND SURFACE WATER - DRAINS AND OIL SEPARATOR	PLANNED - STAGE 3
25	P-004	TOILETS - ENLARGED PLANS AND ISOMETRICS	PLANNED - STAGE 3
26	P-005	SANITARY DETAILS - SEPTIC TANK, SEPARATOR, MANHOLES	PLANNED - STAGE 3
27	S-001	STRUCTURAL NOTES - MATERIALS, LOADS, SEISMIC DATA	PLANNED - STAGE 4
28	S-002	BUILDINGS FOUNDATIONS - FOOTINGS AND TIE BEAMS	PLANNED - STAGE 4
29	S-003	BUILDINGS FRAME - COLUMNS, BEAMS, ROOF SLAB, STAIR	PLANNED - STAGE 4
30	S-004	TANK SADDLES - SADDLES AND FOUNDATIONS	PLANNED - STAGE 4
31	S-005	BUND WALLS - LIQUID-RETAINING RC WALLS	PLANNED - STAGE 4
32	S-006	BURIED STRUCTURES - SEPTIC TANK, SEPARATOR, FIRE TANK	PLANNED - STAGE 4
33	S-007	SITE STRUCTURES - BOUNDARY WALL, PIERS, PLINTHS	PLANNED - STAGE 4
34	S-008	CANOPY FOUNDATIONS - AFTER ARCHITECTURAL CANOPY DESIGN	HOLD - CANOPY DESIGN
35	S-009	CANOPY STEEL STRUCTURE - AFTER ARCHITECTURAL CANOPY DESIGN	HOLD - CANOPY DESIGN

STAGE 6 (S-008, S-009 AND THE CANOPY PARTS OF E-004, E-008, E-010, P-003) STARTS AFTER THE ARCHITECTURAL 3D ELEVATIONS AND THE CANOPY FORM ARE APPROVED.

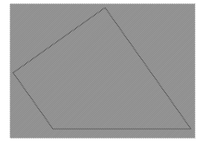
GENERAL

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GENERAL NOTES

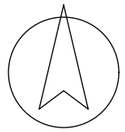
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KEY PLAN



THIS SHEET

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TO BE CONFIRMED

REVISIONS

REV	DATE	DESCRIPTION	BY
00	18/10/2026	FIRST ISSUE - STAGE 0	I.M.

PROJECT
FUEL FILLING STATION
PLOT 1/3453, DISTRICT 43 YARIMJA AL-SHARQIYA

LOCATION
MOSUL, NINEVEH, IRAQ

CLIENT / OWNER
HADI HUSSEIN JUMAA - AHMED OTHMAN EID

PROJECT No.
FS-3453-43

DRAWING TITLE
COVER SHEET
DRAWING REGISTER

ROLE	NAME	SIGN.	DATE
DESIGNED	IBRAHIM MAHFOUDH		18/10/26
DRAWN	IBRAHIM MAHFOUDH		18/10/26
CHECKED	-		18/10/26
APPROVED	-		18/10/26

SCALE
NTS

DATE
18/10/2026

PHASE
DESIGN DEVELOPMENT

STATUS
FOR REVIEW

DRAWING No.
G-001

SHEET
01
OF 3

CIVIL DEFENCE

SYMBOL	CODE	DESCRIPTION	الوصف
P	FE-P	PORTABLE EXTINGUISHER, DRY POWDER 12 kg	مطفأة بودرة جافة 12 كغم
C	FE-C	PORTABLE EXTINGUISHER, CO2 6 kg	مطفأة ثاني اوكسيد الكاربون 6 كغم
F	FE-F	PORTABLE EXTINGUISHER, AFFF FOAM 9 L	مطفأة رغوة 9 لتر
50	FE-W	WHEELED EXTINGUISHER, DRY POWDER 50 kg	مطفأة بودرة جافة بعجلات 50 كغم
	FH	FIRE HYDRANT (2 x 65 mm OUTLETS)	حنفية حريق
HR	HR	FIRE HOSE REEL 25 mm x 30 m	بكرة خرطوم حريق
FACP	FACP	FIRE ALARM CONTROL PANEL	لوحة سيطرة انذار الحريق
S	SD	SMOKE DETECTOR	كاشف دخان
H	HD	HEAT DETECTOR	كاشف حرارة
MCP	MCP	MANUAL CALL POINT	نقطة انذار يدوية
B	SNDR	ALARM SOUNDER / FLASHER	جرس وضوء انذار
ESD	ESD	EMERGENCY SHUT-DOWN PUSH BUTTON	زر ايقاف الطوارئ
EXIT	EXIT	EMERGENCY EXIT SIGN (ILLUMINATED)	علامة مخرج طوارئ
AP	AP	ASSEMBLY POINT	نقطة تجمع
FP	FP	FIRE PUMP SET	مضخة حريق
	WS	WATER SPRAY NOZZLE (LPG TANK COOLING)	مرش ماء لتبريد خزان الغاز

STRUCTURAL

SYMBOL	CODE	DESCRIPTION	الوصف
	GRID	GRID LINE BUBBLE	محور انشائي
	COL	RC COLUMN	عمود خرساني
	FTG	ISOLATED FOOTING	اساس منفرد
	SEC	SECTION MARK	علامة مقطع

ELECTRICAL

SYMBOL	CODE	DESCRIPTION	الوصف
	TRF	TRANSFORMER 11/0.4 kV	محولة 11/0.4 ك.ف
G	GEN	STANDBY DIESEL GENERATOR	مولدة احتياط
ATS	ATS	AUTOMATIC TRANSFER SWITCH	مفتاح تحويل اوتوماتيكي
	MDB	MAIN / SUB DISTRIBUTION BOARD	لوحة توزيع رئيسية / فرعية
	LP	LED PANEL 600x600, 36 W	وحدة انارة LED
	LEX	Ex-d LED LUMINAIRE (HAZARDOUS AREA)	وحدة انارة مقاومة للانفجار
EX	POLE	YARD LIGHTING POLE, LED FLOODLIGHT	عمود انارة ساحة
	EM	EMERGENCY LIGHT, 3 h BATTERY	انارة طوارئ
	SO	SWITCHED SOCKET OUTLET 13 A	مأخذ قدرة 13 امبير
	SW	LIGHTING SWITCH	مفتاح انارة
	JB	JUNCTION BOX	صندوق توصيل
	EP	EARTH PIT WITH TEST LINK	بئر تأريض
AT	AT	LIGHTNING AIR TERMINAL	مانعة صواعق

SANITARY

SYMBOL	CODE	DESCRIPTION	الوصف
	MH	MANHOLE / INSPECTION CHAMBER	منهول / غرفة تفتيش
	GT	GULLY TRAP	مصيدة مياه
	FD	FLOOR DRAIN	بالوعة ارضية
CO	CO	CLEAN-OUT	فتحة تنظيف
M	WM	WATER METER	مقياس ماء
	GV	GATE VALVE	صمام بوابي
	NRV	NON-RETURN VALVE	صمام عدم رجوع
	PU	BOOSTER PUMP	مضخة رفع

LINE TYPES

LINE	LAYER	DESCRIPTION	الوصف
	F-PIPE	FIRE WATER MAIN (HDPE PN16 BURIED)	خط ماء الحريق
	E-POWR	BURIED POWER CABLE IN TRENCH / DUCT	كابل قدرة مدفون
	E-GRND	BARE COPPER EARTH CONDUCTOR 70 mm2	موصل تأريض نحاسي
	E-HAZD	HAZARDOUS AREA BOUNDARY (ZONE 1 / 2)	حدود المنطقة الخطرة
	P-WATR	COLD WATER SUPPLY (PPR)	خط ماء بارد
	P-SANR	FOUL DRAIN (uPVC)	خط صرف صحي
	P-STRM	OILY / SURFACE WATER DRAIN	خط تصريف مياه سطحية / زيتية

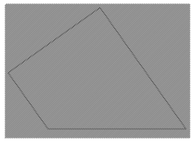
GENERAL

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GENERAL NOTES

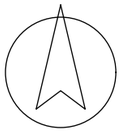
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KEY PLAN



THIS SHEET

NORTH N



TO BE CONFIRMED

REVISIONS

REV	DATE	DESCRIPTION	BY
00	18/10/2026	FIRST ISSUE - STAGE 0	I.M.

PROJECT
FUEL FILLING STATION
PLOT 1/3453, DISTRICT 43 YARIMJA AL-SHARQIYA

LOCATION
MOSUL, NINEVEH, IRAQ

CLIENT / OWNER
HADI HUSSEIN JUMAA - AHMED OTHMAN EID

PROJECT No.
FS-3453-43

DRAWING TITLE
LEGEND
SYMBOLS AND LINE TYPES

ROLE	NAME	SIGN.	DATE
DESIGNED	IBRAHIM MAHFOUDH		18/10/26
DRAWN	IBRAHIM MAHFOUDH		18/10/26
CHECKED	-		18/10/26
APPROVED	-		18/10/26

SCALE	DATE
NTS	18/10/2026
PHASE: DESIGN DEVELOPMENT	STATUS: FOR REVIEW

DRAWING No.
G-002

SHEET
02
OF 3

PROJECT DATA

SITE

PLOT AREA 2,415.25 m2 (RIGHT TRAPEZOID 50 x 30.59 / 66.02)

STREET FRONT 61.28 m, ONE EXIT 10 m, ONE ENTRY 10 m, YARD GATE 5 m

PRODUCTS AND STORAGE (ALL TANKS ABOVE GROUND IN RC BUNDS)

GAS OIL 1 TANK(S) 54 m3

LPG 2 TANK(S) 46 m3

GASOLINE 2 TANK(S) 108 m3

KEROSENE 1 TANK(S) 54 m3

TOTAL 262 m3 - BUND COMPARTMENTS C1 (GAS OIL), C2 (LPG), C3 (PETROL, KEROSENE)

DISPENSING

I-1 GAS OIL 2 DISPENSER(S)

I-2 LPG 2 DISPENSER(S)

I-3 GASOLINE 2 DISPENSER(S)

I-4 KEROSENE (YARD) 2 DISPENSER(S)

I-5 KEROSENE (HAND CARTS) 1 DISPENSER(S)

UTILITIES

POWER: 11 kV SUPPLY, DEDICATED 11/0.4 kV TRANSFORMER IN THE TRANSFORMER ROOM, STANDBY DIESEL GENERATOR WITH AUTOMATIC TRANSFER SWITCH.

WATER: MUNICIPAL MAINS + GROUND STORAGE TANK + BOOSTER PUMP.

DRAINAGE: NO PUBLIC SEWER - SEPTIC TANK AND SOAKAWAY; OILY WATER THROUGH AN OIL SEPARATOR BEFORE DISPOSAL.

DESIGN ASSUMPTIONS (TO BE CONFIRMED)

ALLOWABLE SOIL BEARING 100 kN/m2 PENDING A SOIL INVESTIGATION.

BUILDINGS: REINFORCED CONCRETE FRAME AND SOLID ROOF SLAB.

CANOPY: FORM AND STRUCTURE PENDING ARCHITECTURAL DESIGN (STAGE 6).

NORTH POINT NOT SHOWN ON THE SOURCE PLAN - TO BE CONFIRMED ON SITE.

GENERAL NOTES

1. THIS PACKAGE IS BASED ON THE PRELIMINARY SITE PLAN (FUEL_STATION_SITE_PLAN.dxf) REDRAWN FROM ITS WRITTEN DIMENSIONS. ANY CHANGE TO THE SITE PLAN IS TO BE REFLECTED IN ALL DISCIPLINES.
2. ALL SAFETY DISTANCES ARE CHECKED ON F-001; ANY SHORTFALL IS REPORTED, NOT HIDDEN.
3. ALL ELECTRICAL EQUIPMENT INSIDE ZONE 1 / ZONE 2 SHALL BE CERTIFIED FOR THE ZONE (Ex d / Ex e, IIA T3 MINIMUM, IIB FOR LPG AREAS).
4. ALL CALCULATED VALUES SHOWN ON THE DRAWINGS COME FROM THE DESIGN REPORT (calcs.py).
5. THE DRAWINGS ARE FOR AUTHORITY SUBMISSION AND REVIEW AND REQUIRE THE SIGNATURE OF A REGISTERED ENGINEER (IRAQI ENGINEERS UNION).
6. CANOPY-DEPENDENT ITEMS ARE ON HOLD UNTIL THE ARCHITECTURAL CANOPY DESIGN IS APPROVED (STAGE 6).

IRAQI CODES AND AUTHORITY REQUIREMENTS

IRAQI BUILDING CODES (IRAQI BUILDING COUNCIL - MINISTRY OF CONSTRUCTION AND HOUSING):

مدونة التأسيسات الكهربائية - الطبعة الاولى 2016

المواصفات الفنية للأعمال الكهربائية 2015

مدونة التأسيسات المائية في المباني

المواصفات الفنية للأعمال الصحية

مدونات الاحمال والخرسانة المسلحة والزلازل العراقية

CIVIL DEFENCE DIRECTORATE REQUIREMENTS (مديرية الدفاع المدني).

OIL PRODUCTS DISTRIBUTION COMPANY (MINISTRY OF OIL):

INSTRUCTIONS FOR PRIVATE FUEL FILLING STATIONS - LATEST EDITION TO BE OBTAINED; CLAUSES MARKED (OPDC) ARE TO BE VERIFIED AGAINST IT.

INTERNATIONAL STANDARDS

NFPA 30 / NFPA 30A FLAMMABLE LIQUIDS / MOTOR FUEL DISPENSING

NFPA 58 LIQUEFIED PETROLEUM GAS CODE

NFPA 10 / 15 / 72 EXTINGUISHERS / WATER SPRAY / FIRE ALARM

NFPA 70 ART. 514 MOTOR FUEL DISPENSING FACILITIES

IEC 60079-10-1 HAZARDOUS AREA CLASSIFICATION

EI MODEL CODE PART 15 AREA CLASSIFICATION (PETROLEUM)

IEC 60364 / IEC 62305 LV INSTALLATIONS / LIGHTNING PROTECTION

ACI 318 / ACI 350 STRUCTURAL CONCRETE / LIQUID-RETAINING

IPC INTERNATIONAL PLUMBING CODE (REFERENCE)

INFORMATION REQUIRED FROM THE AUTHORITIES / CLIENT

1. OPDC INSTRUCTIONS (LATEST EDITION) - SEPARATION DISTANCES, TANK AND DISPENSER REQUIREMENTS.
2. CIVIL DEFENCE DIRECTORATE - FIRE WATER STORAGE AND HYDRANT REQUIREMENTS FOR THE STATION CATEGORY.
3. ELECTRICITY DIRECTORATE - 11 kV CONNECTION POINT AND TRANSFORMER RATING.
4. MUNICIPAL WATER CONNECTION POINT AND PRESSURE.
5. SOIL INVESTIGATION REPORT (BEARING CAPACITY, GYPSUM CONTENT, WATER TABLE).
6. TRUE NORTH AND SITE LEVELS (TOPOGRAPHIC SURVEY).
7. ARCHITECTURAL CANOPY FORM, HEIGHT AND ELEVATIONS.

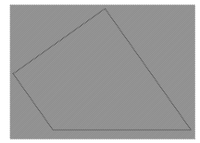
GENERAL

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GENERAL NOTES

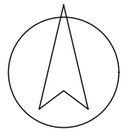
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KEY PLAN



THIS SHEET

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TO BE CONFIRMED

REVISIONS

REV	DATE	DESCRIPTION	BY
00	18/10/2026	FIRST ISSUE - STAGE 0	I.M.

PROJECT
FUEL FILLING STATION
PLOT 1/3453, DISTRICT 43 YARIMJA AL-SHARQIYA

LOCATION
MOSUL, NINEVEH, IRAQ

CLIENT / OWNER
HADI HUSSEIN JUMAA - AHMED OTHMAN EID

PROJECT No.
FS-3453-43

DRAWING TITLE
DESIGN BASIS
CODES, DATA AND GENERAL NOTES

ROLE	NAME	SIGN.	DATE
DESIGNED	IBRAHIM MAHFOUDH		18/10/26
DRAWN	IBRAHIM MAHFOUDH		18/10/26
CHECKED	-		18/10/26
APPROVED	-		18/10/26

SCALE
NTS

DATE
18/10/2026

PHASE
DESIGN DEVELOPMENT

STATUS
FOR REVIEW

DRAWING No.
G-003

SHEET
03
OF 3